

Case Study of Protein Stability by Using Topological Properties *

Jiunn-I Shieh and Kuei-Jen Lee

Department of Information Management
Taichung Healthcare and Management University, Taichung, Taiwan
Department of Bioinformatics
Taichung Healthcare and Management University, Taichung, Taiwan

{jishieh, kjlee}@thmu.edu.tw

Abstract

This paper shows how to study the protein stability by using the topological properties. In our case study, this paper also shows that protein systems have power-law property. This means some atoms are very important to each protein system. We conjecture that there exists a binding site around them.

1 Introduction

Thermostability is one of the most important properties of enzymes in both science and industry. Thermostable proteins can be used to improve the efficiency of chemical processes in industry and provide a better understanding protein folding and stabilization, which are important for protein engineering, biomolecular design, and structural prediction[1]. In this work, we are focused on industrial enzymes which can hydrolyze the cellulose to glucose and is called cellulases.

Cellulases are glycoside hydrolases which can be found in bacteria and fungi. Bacterial and fungal cellulases have been widely used in detergent, textile, and food industries[2]. The enzymes that hydrolyze the cellulose to glucose can be divided in two classes[2]: endoglucanases (EC 3.2.1.4), which hydrolyze internal bonds in the cellulose chains, and exoglucanases (EC 3.2.1.91), which hydrolyze from the chain ends. Both of these enzymes are important for industry, due to the fact that the cellulose can be hydrolysed to glucose by these enzymes and the glucose is fermented to ethanol by yeast for the use of transportation fuel. Cellulases are glycoside hydrolases. In this study, we used glycoside hydrolase family 12 (Cel12) endoglucanases[3] as the studying materials, which are enzymes isolated from *Humicola grisea* (H. grisea Cel12A) and *Trichoderma reesei* (T. reesei Cel12A). H. grisea enzyme has a thermal denaturation midpoint of the transition temperature (T_m) of 68.7°C which is 14.3 °C higher than that of T. reesei enzyme[3]. The H. grisea enzyme has 43% sequence identity

* Research supported by NSC 92-2213-E-468-007

with the *T. reesei* enzyme by the global alignment[4], as shown in Fig.1. The three-dimensional (3-D) structures of the *H. grisea* enzyme and *T. reesei* enzyme have been obtained by X-ray diffraction and are now available in the Protein Data Bank (PDB) with PDB codes 1OLR (Fig.2) and 1H8V (Fig. 3), respectively.

This paper proposes a methodology of using the topological invariant to discriminate two different function proteins although they have high similarity. In Section 2, the definitions of topological invariants are introduced. A novel approach of topological properties is presented in Section 3. Section 4 presents the results of computation and compares the results with other approach. Finally, a conclusion is in the last section.

2 Topological invariants

A topological property is very important for classification in many aspects. The purpose of this paper is to find some good topological properties to classify protein function.

The topological properties (e.g., patterns of connectivity, topological complexity, cycle rank, etc) of graphs can be analyzed using spectral graph theory. Each graph has a Laplacian matrix

representation and a corresponding eigenvalue spectrum [6]; the number of eigenvalues is equal to the rank of the matrix. The eigenvalues are related to the connectivity pattern (or topology) of the graph. In particular, the second largest eigenvalue is a measure of the compactness of a graph. A large second eigenvalue indicates a compact graph, whereas a small eigenvalue implies an elongated topology[5].

Define the Laplacian matrix of a graph $L(G)$.

The Laplacian matrix marks the connections between nodes, such that $L(i, j) = -1$ if $i \neq j$ and i is adjacent to j . Also, $L(i, i) =$ the total number of connections between i and any other node. Thus the Laplacian matrix is both symmetric, and the values of all rows and columns add up to zero. It is known that the second eigenvalue of the Laplacian matrix of a graph, $\lambda_2(L(G))$, is the algebraic connectivity of graph G . Thus the vector $\mathbf{V}_2$ corresponding to $\lambda_2(L(G))$ is the eigenvector of algebraic connectivity, and has special properties.

Another topological property is “cycle rank”.

Let G be a graph with $c(G)$ components. The cycle rank of G , denoted $\beta(G)$, is the number of edges in the relative complement of a full spanning forest of G .

Thus, the cycle rank is $\beta(G) = |E(G)| - |V(G)| + c(G)$, where $|E(G)|$ is the number of edges in the graph G and $|V(G)|$ is the number of vertices in the graph G .

From the definition, it is easy to see that $\beta(G)$ is the minimum number of edges to be deleted for the induce subgraph being a spanning tree. This should be related to thermal property.

In this work the Laplacian matrix is used to calculate its corresponding eigenvalues. The total number of vertices in a protein secondary structure equals the total number of eigenvalues. The eigenvalue that helps to describe the protein topology is the second eigenvalue. The second eigenvalue describes the compactness of a graph. We believe that the compactness of a graph is also related to thermal property

3 Methods

First, find two proteins which have high similarity but different function. We check it by the global alignment. Then, we take the following processes:

- Find 3-dimension coordinates of all atoms in each protein.

- Construct a graph from the above. First, we take the atoms as the vertices of the graph. Then we get a trival graph with empty edge set. Given the threshold $\varepsilon > 0$, we add a edge between two atoms if the distance between the two atoms is greater ε , otherwise no edge is added between two atoms.

- Change the threshold from 4 unit to 30 unit by increasing 1 unit each time. Each time we get a graph(G). Note that the unit is 10^{-8} cm.

- For each graph we write down the Laplacian matrix $L(G)$ of the correspondent graph.

- For each Laplacian matrix, calculate the second eigenvalue $\lambda_2(L(G))$ of the Laplacian

matrix and $\beta(G)$ of the correspondent graph.

4 Data analysis

The data analysis is divided into three cases:

- Case 1: $L_2(L(G))$

From the computation results (Table 4), at many different scales we have that the value L_2 of H. grisea enzyme is greater than that of T. reesei enzyme. So the graph of H. grisea enzyme is more compact than that of T. reesei enzyme. Therefore, it is very reasonable to say that the thermal stability of H. grisea enzyme is higher than that of T. reesei enzyme.

- Case 2: $\beta(G)$.

From the computation results (Table 5), at many different scales we have that the value cycle rank of H. grisea enzyme is greater than that of T. reesei enzyme. It means that the graph of H. grisea enzyme needs more time to delete the edge to form a spanning tree than that of H. grisea enzyme. Therefore, it is also reasonable to say that the thermal stability of H. grisea enzyme is higher than that of T. reesei enzyme.

- Case 3:

From Table 4 and Table 5, we check the power-law property for each system by log-log regression. The results show that r^2 for L_2 of T. reesei enzyme and H. grisea enzyme are 0.992951 and 0.992849 as shown in Table 6 and Table 7, respectively. The r^2 for $\beta(G)$ of T. reesei enzyme and H. grisea enzyme are 0.98586 and 0.98708 as shown in Table 8 and Table 9, respectively. These show that each system is very good fit to power-law property.

To sum up, compare L_2 method and $\beta(G)$ method from data we analyzed. It seems that $\beta(G)$ method is better than L_2 method.

5 Conclusion

In this paper, we have shown that the topological properties can be used to discriminate two different function proteins although they have high similarity. These two protein system also have power-law property. This means that some atoms are very important to each protein system. We conjecture that there maybe a binding site around them. Currently, we are still working on

the issues of the study of protein stability by using topological properties within a shorter time and the algorithm for finding the binding sites.

References

- [1] Kim, Y.-W. et al., 2003, Directed Evolution of Thermus Maltogenic Amylase toward Enhanced Thermal Resistance, Appl. Environ. Microbiol. 8, 4866-4874.
- [2] Schulein, M, 2000, Protein engineering of cellulases, Biochimica et Biophysica Acta, 1543, 239-252.
- [3] Sandgren, M. et al., 2003, The Humicola grisea Cel12A enzyme structure at 1.2 Å resolution and the impact of its free cysteine residues on thermal stability, Protein

Science, 12, 2782-2793.

- [4] <http://www.ebi.ac.uk/emboss/align/>
- [5] Hin Hark Gan, Samuela Pasquali, and Tamar Schlick, Exploring the repertoire of RNA secondary motifs using graph theory with implications for RNA design, Nucleic Acids Res., 31, 2926-2943, 2003.
- [6] F.R.K. Chung, Spectral graph theory, Amer. Math. Soc. Providence, 1994.

Appendix

```
#=====
#
# Aligned_sequences: 2
# 1: 10LR
# 2: 1H8V
# Matrix: EBLOSUM62
# Gap_penalty: 10.0
# Extend_penalty: 0.5
#
# Length: 229
# Identity: 98/229 (42.8%)
# Similarity: 137/229 (59.8%)
# Gaps: 16/229 ( 7.0%)
# Score: 501.5
#
#
#=====
```

10LR	1	XIRSLCELYGYWSGNGYELLNNLWGKDTATSGWQCTYLDGTNNGGIQWST	50
1H8	1	. . . : : : . : : . . .	46

[illegible]

Fig. 1. Global alignment of *H. grisea* enzyme (1OLR) and *T. reesei* enzyme (1H8V).

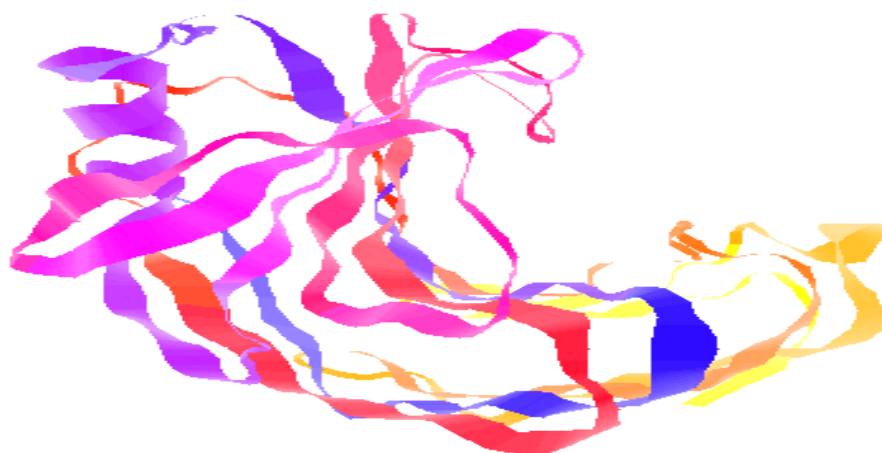


Fig. 2. Three-dimensional structure of the H. grisea enzyme

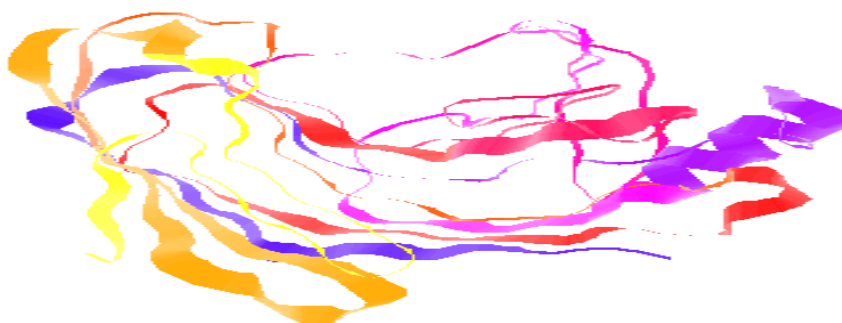


Fig. 3. Three-dimensional structure of the T. reesei enzyme.

ε	l_2 (T. reesei enzyme)	l_2 (H. grisea enzyme)		ε	Cycle rank (T. reesei enzyme)	Cycle rank (H. grisea enzyme)
4	0.0720	0.0724		4	8364	9284
5	0.2835	0.2835		5	18517	20602
6	0.7070	0.7250		6	32111	36024

7	1.5495	1.5812		7	49337	55492
8	2.9851	3.0645		8	69818	78801
9	5.3870	5.5512		9	95002	107423
10	9.2055	9.4029		10	125734	142228
11	14.8544	15.2237		11	161559	183373
12	22.7464	23.1884		12	201092	228431
13	33.1815	33.8253		13	243904	278035
14	47.0273	47.7713		14	291456	332716
15	64.9195	65.7445		15	343658	392409
16	86.7573	88.0187		16	397901	455325
17	113.0886	114.8249		17	453930	520824
18	144.2712	146.3088		18	511943	588210
19	180.4478	183.1114		19	571833	658189
20	222.1353	225.4988		20	632657	729787
21	268.4324	272.3763		21	693098	801556
22	318.9149	323.7338		22	752409	872813
23	374.5746	379.7790		23	811528	943894
24	434.5988	440.5198		24	869608	1014301
25	497.5034	504.7836		25	925416	1082854
26	563.8181	572.2726		26	978823	1148567
27	632.0749	641.4141		27	1028911	1210960
28	701.3060	713.3066		28	1076467	1270850
29	766.0013	787.7856		29	1120604	1327040
30	839.9631	858.7090		30	1160743	1378806
31	915.2513	919.9922		31	1197074	1425901

Table 4. l_2

Table5. β (G) (cycle rank)

SUMMARY

OUTPUT

<i>Regression Statistics</i>	
Multiple R	0.996469
R Square	0.992951

Adjusted R Square	0.99268
Standard Error	0.324813
Observations	28

ANOVA

	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>	
Regression	1	386.3821	386.3821	3662.276	1.65E-29	
Residual	26	2.743085	0.105503			
Total	27	389.1252				

	<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>	<i>Lower 95%</i>	<i>Upper 95%</i>
Intercept	-12.1091	0.3021	-40.083	6.62E-25	-12.7301	-11.4881
X variable 1	4.553313	0.075241	60.51674	1.65E-29	4.398654	4.707973

Table 6. $\log_2(I_2) - \log_2 e$ (T. reesei enzyme)

SUMMARY

OUTPUT

<i>Regression Statistics</i>	
Multiple R	0.996418
R Square	0.992849
Adjusted R Square	0.992574
Standard Error	0.327155
Observations	28

ANOVA

	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>	
Regression	1	386.3547	386.3547	3609.767	1.99E-29	
Residual	26	2.78279	0.10703			
Total	27	389.1375				

	<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>	<i>Lower 95%</i>	<i>Upper 95%</i>
Intercept	-12.0844	0.304279	-39.715	8.38E-25	-12.7099	-11.459
X variable 1	4.553152	0.075783	60.08133	1.99E-29	4.397377	4.708926

Table 7. $\log_2(I_2) - \log_2 e$ (H. grisea enzyme)

SUMMARY

OUTPUT

Regression Statistics						
Multiple R	0.992905					
R Square	0.98586					
Adjusted R Square	0.985316					
Standard Error	0.238165					
Observations	28					
ANOVA						
	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>	
Regression	1	102.8228	102.8228	1812.728	1.41E-25	
Residual	26	1.47479	0.056723			
Total	27	104.2976				
	<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>	<i>Lower 95%</i>	<i>Upper 95%</i>
Intercept	8.976274	0.221512	40.5228	5E-25	8.52095	9.431598
X variable 1	2.348896	0.055169	42.57614	1.41E-25	2.235494	2.462298

Table 8. $\log_2(\beta(G)) - \log_2 \varepsilon$ (T. reesei enzyme)

SUMMARY

OUTPUT

<i>Regression Statistics</i>						
Multiple R	0.993519					
R Square	0.98708					
Adjusted R Square	0.986583					
Standard Error	0.230687					
Observations	28					

ANOVA

	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance</i> <i>F</i>	
Regression	1	105.704	105.704	1986.307	4.36E-26	
Residual	26	1.383625	0.053216			
Total	27	107.0876				
	<i>Coefficients</i>	<i>Standard</i> <i>Error</i>	<i>t Stat</i>	<i>P-value</i>	<i>Lower 95%</i>	<i>Upper 95%</i>
Intercept	9.049588	0.214556	42.17821	1.79E-25	8.608561	9.490614
X variable 1	2.381577	0.053437	44.56801	4.36E-26	2.271736	2.491419

Table 9. $\log_2(\beta(G))$ -- $\log_2 \varepsilon$ (H. grisea enzyme)